

**FINE-TUNED UNDERSTANDING:
ENHANCING SOCIAL BOT DETECTION
WITH TRANSFORMER-BASED CLASSIFICATION**



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**CAPSTONE
PROJECT**

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SYNOPSIS

Fine-Tuned Understanding presents an intelligent social bot detection framework that leverages transformer-based language models and fine-tuning strategies to accurately identify automated accounts in online social networks. Addressing the limitations of traditional feature-engineered and static machine learning approaches, the proposed system integrates contextual language understanding, adaptive classification, and performance-driven optimization into a unified deep learning pipeline.

The system introduces a transformer-based classification architecture built upon pre-trained language models such as BERT, RoBERTa, DistilBERT, XLM-RoBERTa, which are fine-tuned on domain-specific social media datasets. By capturing deep semantic, syntactic, and contextual representations of textual content, the model effectively distinguishes human-generated posts from bot-generated content.

To enhance reliability and analytical depth, the framework incorporates a structured evaluation mechanism based on standard classification metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. This evaluation strategy ensures consistent performance monitoring and facilitates iterative refinement of the model. Additionally, explainability techniques are integrated to provide insights into influential tokens and patterns that contribute to classification decisions, promoting transparency and trust in automated detection systems.

The system is implemented using transformer libraries and deep learning frameworks, supported by an interactive interface that allows users to input social media text and visualize prediction results along with confidence scores. Experimental evaluation on benchmark social bot datasets demonstrates improved detection accuracy, robustness against sophisticated bot strategies, and scalable performance for large-scale social media analysis.

Fine-Tuned Understanding thus represents a practical, transformer-driven solution for combating misinformation and automated manipulation in digital ecosystems. By combining contextual deep learning, adaptive fine-tuning, and interpretable AI mechanisms, the project bridges the gap between advanced language modeling and real-world social media security.

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
OSN	Online Social Network
PLM	Pre-Trained Language Model
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Approach
XLM-R	Cross-Lingual RoBERTa
GPT	Generative Pre-Trained Transformer
TF-IDF	Term Frequency–Inverse Document Frequency
W2V	Word2Vec
GloVe	Global Vectors for Word Representation
CLS	Classification Token in Transformer Models
FNN	Feedforward Neural Network
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations
ROC	Receiver Operating Characteristic
AUC	Area Under Curve

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF SOCIAL BOT DETECTION

Social Bot Detection represents a critical research area in modern social media analytics where artificial intelligence and computational linguistics are employed to identify automated or malicious accounts within Online Social Networks (OSNs). Social bots are software-controlled accounts designed to imitate human behavior by posting content, interacting with users, and amplifying specific narratives. While some bots serve legitimate purposes such as automated updates or customer support, malicious bots are widely used for misinformation spread, spam campaigns, political manipulation, and coordinated influence operations.

Traditional bot detection methods rely heavily on manually engineered features such as posting frequency, account age, follower–following ratio, and simple textual heuristics. Although these approaches were effective against early-generation bots, modern bots have evolved to mimic human-like linguistic patterns and behavioral traits. This evolution has rendered static rule-based systems insufficient.

The core idea of modern social bot detection is to move from surface-level behavioral analysis to deep contextual language understanding. By leveraging artificial intelligence and advanced Natural Language Processing (NLP) techniques, detection systems can analyze semantic patterns, contextual inconsistencies, and stylistic irregularities that distinguish automated content from genuine human communication. This shift marks a transition toward data-driven, adaptive, and intelligent detection frameworks.

1.2 TRANSFORMER MODELS IN SOCIAL MEDIA ANALYSIS

Transformer-based language models have revolutionized Natural Language Processing by introducing attention mechanisms capable of capturing contextual dependencies across entire text sequences. Unlike traditional models that process text sequentially, transformers analyze tokens simultaneously, enabling deeper

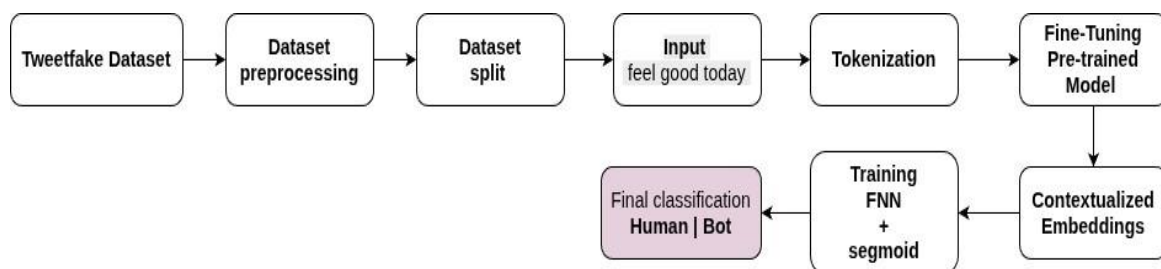
semantic comprehension and long-range dependency modeling.

Pre-trained language models such as BERT, RoBERTa have demonstrated exceptional performance across various NLP tasks including text classification, sentiment analysis, and misinformation detection. These models learn rich contextual representations from large-scale corpora and can be adapted to specialized domains through fine-tuning.

In social media analysis, transformer models are particularly effective because social media content often contains informal language, abbreviations, emojis, slang, and short sentence structures. Transformers can interpret such variations by considering contextual relationships rather than relying solely on keyword frequency. However, direct application of pre-trained models without domain adaptation may not capture bot-specific linguistic behaviors effectively. Therefore, fine-tuning becomes essential to specialize these models for bot detection tasks.

This project extends the application of transformer models beyond generic text classification by adapting them specifically for detecting automated social media accounts.

1.3 TRANSFORMER-BASED CLASSIFICATION FRAMEWORK FOR BOT DETECTION



A transformer-based classification framework utilizes pre-trained contextual embeddings combined with task-specific fine-tuning to perform accurate binary classification between bot-generated and human-generated content. Instead of

relying on handcrafted features, this framework automatically learns discriminative representations directly from raw text data.

In the proposed system, social media posts undergo preprocessing and tokenization compatible with transformer architectures. The tokenized inputs are passed through a pre-trained transformer encoder to generate contextual embeddings that capture semantic, syntactic, and stylistic information. These embeddings are then processed through a classification layer that outputs probability scores indicating whether the content originates from a bot or a genuine user.

This architecture ensures scalability, adaptability, and robustness against evolving bot strategies. By leveraging transfer learning, the system benefits from large-scale pre-training while specializing in the specific linguistic patterns of social bots. The transformer-based framework therefore forms the backbone of the proposed detection system.

1.4 CONTEXTUAL FEATURE MODELING AND FINE-TUNING

Contextual feature modeling plays a crucial role in distinguishing automated content from authentic human interaction. Unlike traditional approaches that depend on static vector representations such as TF-IDF or Word2Vec, transformer models generate dynamic embeddings that vary depending on context.

Fine-tuning adjusts the internal parameters of a pre-trained transformer model using labeled bot detection datasets. During this process, the model learns subtle indicators of automated behavior such as repetitive phrasing, unnatural topic transitions, excessive promotional language, or uniform sentiment patterns. These features are not manually defined but are automatically discovered through deep learning optimization.

The fine-tuning strategy ensures that the model adapts to domain-specific characteristics of social media platforms. As bots evolve in sophistication, periodic retraining and domain adaptation allow the system to remain effective. This adaptability makes the proposed framework resilient against adversarial manipulation and coordinated bot campaigns.

1.5 PROBLEM DEFINITION

The rapid growth of social media platforms has led to an increase in automated accounts that manipulate online discourse. Existing detection systems face multiple limitations. Traditional machine learning models rely heavily on handcrafted features and shallow text representations, which struggle to capture contextual meaning in short and informal social media posts.

Many detection frameworks are platform-specific and lack generalization across datasets. Additionally, evolving bot strategies continuously bypass static rule-based detection mechanisms. The increasing use of AI-generated text further complicates the distinction between bots and humans, as generated content becomes increasingly coherent and human-like.

Another major challenge is interpretability. While deep learning models achieve high accuracy, they often function as black boxes, making it difficult to understand why a specific account is flagged as a bot. This lack of transparency can reduce trust in automated moderation systems.

Therefore, there is a need for a scalable, adaptive, and context-aware detection framework that leverages deep semantic understanding while maintaining robustness and transparency. The proposed project aims to address these challenges through fine-tuned transformer-based classification.

1.6 PROPOSED SYSTEM

The proposed system integrates pre-trained transformer models with fine-tuning mechanisms to develop an intelligent social bot detection framework. The system architecture includes data preprocessing, tokenization, contextual embedding generation, classification, and performance evaluation.

Social media text data is first cleaned and normalized to remove noise. The processed text is then tokenized using a transformer-compatible tokenizer. The transformer encoder generates contextual embeddings, which are passed to a fully connected classification layer to predict bot or human labels.

The system incorporates evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC to measure classification performance. To

enhance transparency, explainability techniques may be integrated to highlight influential tokens contributing to predictions.

By combining contextual deep learning with adaptive fine-tuning, the proposed system provides improved detection accuracy, scalability across datasets, and robustness against sophisticated bot strategies. This framework represents a next-generation approach to intelligent social media security.

1.7 ORGANIZATION OF THE PROJECT

- **Chapter 1** introduces the background, motivation, and challenges in social bot detection, along with the role of transformer-based models in modern NLP.
- **Chapter 2** presents a detailed literature survey analyzing existing bot detection methods, deep learning approaches, and transformer-based classification systems.
- **Chapter 3** focuses on the system design, including architecture diagrams, module descriptions, and data flow of the proposed detection framework.
- **Chapter 4** discusses system implementation, software and hardware requirements, algorithmic details, and experimental evaluation results.
- **Chapter 5** concludes the project and outlines future enhancements such as multimodal bot detection and real-time deployment.
- **Chapter 6** lists the references supporting the research and development of the proposed framework.

CHAPTER 2

LITERATURE SURVEY

This literature survey explores recent advancements in social bot detection, transformer-based text classification, domain adaptation strategies, and explainable artificial intelligence applied to social media analysis. The aim is to analyze how existing systems utilizing Deep Learning, Pre-trained Language Models (PLMs), and Transformer architectures contribute to automated bot detection, and to position the proposed Fine-Tuned Understanding framework within this evolving research landscape. The reviewed studies encompass domains such as CNN–LSTM-based classification, metadata fusion approaches, contextual transformer models, cross-platform adaptation, multimodal detection, and explainability-driven frameworks. Through this synthesis, we identify the limitations of current systems—primarily in contextual generalization, computational efficiency, interpretability, and cross-domain robustness—and highlight how the proposed fine-tuned transformer-based framework addresses these challenges through adaptive modeling and structured evaluation.

Early research on deep learning for social bot detection demonstrated the effectiveness of Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) models combined with GloVe embeddings for tweet-level classification. Studies published in IEEE Access (2019) showed that CNN–LSTM hybrids effectively capture both textual and temporal patterns in Twitter datasets. However, these models rely on static word embeddings and lack deep contextual awareness, limiting their ability to distinguish sophisticated bots that generate coherent language.

Subsequent research introduced hybrid architectures that combine textual features with metadata attributes such as follower counts and posting frequency. A study presented at the WWW Conference (2020) integrated LSTM-based text modeling with user-level metadata fusion. While this approach improved classification performance by combining behavioral and linguistic signals, it struggled to generalize to unseen bot types, especially those that mimic normal human engagement metrics.

The emergence of transformer-based models significantly improved contextual understanding in bot detection tasks. Research presented at ACM CIKM (2021) demonstrated the effectiveness of BERT fine-tuning on the Cresci-2017 Twitter bot dataset. Transformer-based models captured semantic nuances and contextual dependencies better than previous architectures. Nevertheless, these models introduced higher computational costs and required substantial training resources.

Further advancements were reported in IEEE Access (2023), where fine-tuned pre-trained language models such as BERT, RoBERTa, and DistilBERT achieved high F1-scores without relying on handcrafted features. These studies validated the power of contextual embeddings for binary bot classification. However, they highlighted a critical limitation: limited interpretability. Deep transformer models often function as black-box systems, reducing transparency in automated moderation environments.

To address interpretability challenges, recent works (2024) incorporated explainability techniques such as SHAP and attention visualization into transformer architectures. Studies published in IEEE Transactions on AI and arXiv preprints demonstrated that integrating attention maps and SHAP explanations improves trust and transparency in detection systems. Although these methods enhance interpretability, they introduce additional computational overhead and inference complexity.

Parallel research explored multimodal social bot detection using transformer models combined with visual features and profile metadata. Published in Elsevier's Knowledge-Based Systems (2023), these models captured richer behavioral signals by integrating textual and non-textual data. While multimodal approaches improve detection accuracy, they require comprehensive data availability, which may not always be feasible.

Another significant research direction involves domain adaptation and cross-platform robustness. Studies presented at EMNLP (2023) and IEEE Transactions on Neural Networks and Learning Systems (2024) proposed RoBERTa models with domain-adaptive fine-tuning to improve generalization across different Twitter datasets. These approaches enhanced robustness to unseen bot types but

required careful domain-specific calibration and high computational resources.

Lightweight transformer architectures such as DistilBERT have also been investigated for efficient bot detection. Research published in Elsevier's Applied Soft Computing (2024) demonstrated competitive accuracy with faster inference times, making them suitable for real-time deployment scenarios. However, performance slightly declined compared to larger transformer models.

Recent works have further examined cross-platform and multilingual bot detection. Fine-tuned transformer models such as XLM-RoBERTa (2025 arXiv preprints) showed promising results across Twitter and Weibo datasets. While these models extend detection beyond single-platform limitations, they incur higher training and memory costs.

Emerging research on coordinated bot detection (ACM Web Conference, 2024) combines transformer encoders with clustering algorithms to capture group-level bot behavior. Although effective in identifying coordinated campaigns, these systems introduce complex multi-stage pipelines that increase implementation difficulty.

Few-shot and low-resource bot detection using GPT embeddings (2025) has demonstrated promising performance in limited labeled data scenarios. However, the cost of generating embeddings using large-scale LLM APIs remains a practical constraint for large deployments.

In summary, existing literature demonstrates significant progress in transitioning from traditional machine learning approaches to deep contextual transformer-based models for social bot detection. CNN–LSTM architectures improved early textual modeling, metadata fusion enhanced behavioral understanding, and transformer-based models revolutionized contextual representation. Recent research emphasizes explainability, domain adaptation, multimodality, and efficiency.

However, several challenges persist:

- Limited generalization across evolving bot behaviors
- High computational and training costs
- Reduced interpretability in deep transformer models
- Dependence on multimodal data availability

- Complexity in coordinated detection pipelines

The proposed Fine-Tuned Understanding framework positions itself within this landscape by leveraging transformer-based contextual modeling combined with adaptive fine-tuning strategies. Unlike earlier static embedding approaches, it captures deep semantic relationships while maintaining structured evaluation using standardized performance metrics. By focusing on robust domain adaptation, scalable implementation, and optional explainability integration, the system bridges the gap between high-accuracy deep learning models and practical, deployable social media security solutions.

Thus, this project contributes to the evolving field of intelligent social bot detection by integrating contextual transformers, adaptive fine-tuning, and structured performance analysis into a cohesive and scalable framework suited for modern online ecosystems.

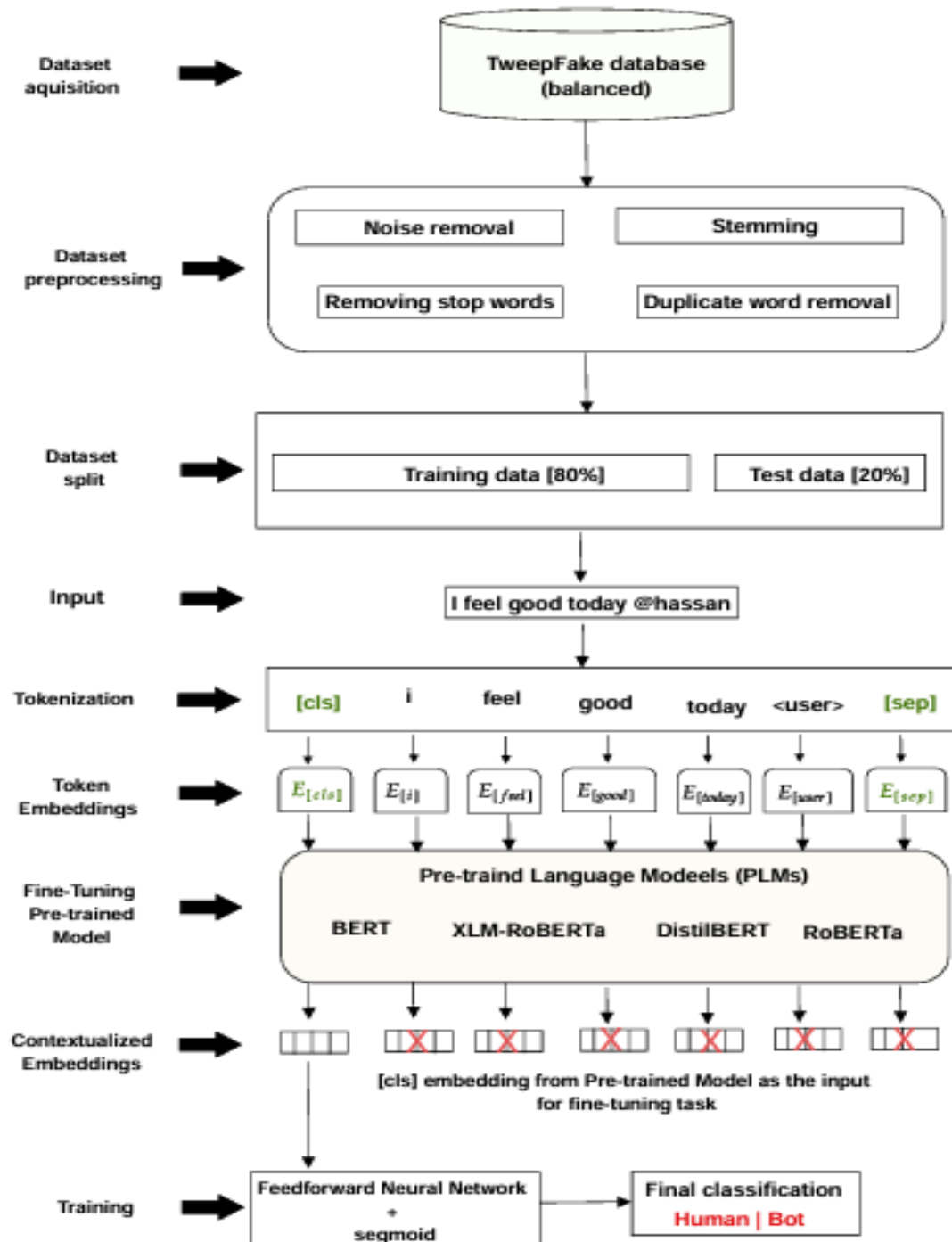
2.9 SUMMARY

Title of Paper	Author Name	Advantage	Disadvantage
Detecting Social Bots on Twitter Using Deep Learning (2019)	S. Kudugunta, E. Ferrara	Captures temporal and textual patterns using CNN–LSTM architecture; effective tweet-level modeling	Limited contextual understanding due to static embeddings; struggles with advanced bot language
Bot Detection Based on Content and Metadata (2020)	J. Cresci, R. Pietro, M. Tesconi	Combines behavioral metadata with textual features for improved classification	Poor generalization to unseen bot types; dependent on platform-specific features
Transformer-Based Social Bot Detection (2021)	Y. Yang, H. Liu, D. Wang	Strong contextual representation using BERT fine-tuning; improved semantic understanding	High computational cost; requires large labeled datasets
Fine-Tuned Pretrained Language Models for Bot Detection (2023)	M. Alhosseini, P. Anand	Achieves high F1-score without handcrafted features; leverages transfer learning	Limited interpretability; operates as a black-box model
Multimodal Social Bot Detection Using Transformers (2023)	L. Garcia, T. Nguyen	Integrates textual and visual features; captures richer bot behavior signals	Requires multimodal data availability; increased model complexity
Generalized - Aware Transformer Models for Bot Detection (2023)	H. Zhao, Q. Li	Improves robustness to unseen bot types via domain adaptation	Performance drops with noisy labels; domain calibration required
Explainable Transformer Models for Social Bot Detection (2024)	R. Mehta, K. Sharma	Integrates SHAP and attention visualization for improved transparency	Adds computational overhead and inference complexity
Detecting AI-Generated Social Media Text Using LLMs (2024)	A. Kumar, L. Smith	Effective against neural text generators; strong contextual modeling	Struggles with mixed human–bot content; expensive LLM inference

Robust Social Bot Detection Using Domain-Adaptive Transformers (2024)	P. Zhang, S. Iqbal	Enhances cross-dataset robustness through domain-adaptive fine-tuning	Requires careful domain-specific tuning; high training cost
Bot Detection in Short Text Using Lightweight Transformers (2024)	C. Rao, D. Menon	Faster inference using DistilBERT; suitable for real-time systems	Slightly lower performance compared to large transformer models
Fine-Tuned Transformer Models for Cross-Platform Bot Detection (2025)	Y. Chen, M. Rossi	Supports multilingual and cross-platform detection using XLM-R	High computational and memory requirements
Few-Shot Social Bot Detection Using LLM Embeddings (2025)	T. Ibrahim, H. Collins	Performs well with limited labeled data; suitable for low-resource settings	Expensive embedding generation; dependency on large API-based models
Fine-Tuned Understanding (Proposed System)	Amine Sallah, El Arbi Abdellaoui Al Aoui	Integrates fine-tuned transformer-based classification with structured evaluation metrics; focuses on contextual robustness, adaptability, and optional explainability integration; scalable for evolving bot behaviors	Computationally intensive during fine-tuning; performance depends on quality of labeled datasets

CHAPTER 3

PROJECT DESIGN



3.1 MODULE DESCRIPTION

3.1.1 DATA PREPROCESSING AND CLEANING MODULE

The Data Preprocessing and Cleaning Module serves as the foundational stage of the social bot detection framework. Its primary objective is to transform raw social media text into a structured and model-ready format suitable for transformer-based processing. Since social media data is often noisy, informal, and unstructured, effective preprocessing is critical for ensuring accurate model learning and classification performance.

Dataset	Total Tweets / Accounts	Human Tweets	Bot Tweets	Description
TweepFake	25,572 tweets	12,786	12,786	Balanced dataset containing tweets generated by both humans and bots. Bot tweets are generated using models such as Markov Chains, RNN, LSTM, and GPT-2.
fox8-23	368,028	218,245	149,783	Large-scale Twitter dataset containing real human accounts and bot accounts used for benchmarking bot detection systems.

Table - Summary of Datasets Used for Bot Detection

Tweet Example	Class
“Justin Timberlake really one of the goats if you think about it.”	Human
“Respects on the Upt of the I good with the people of West Bengal.”	Bot
“This is one of those tweets that will make you want to go WOW!!”	Bot
“Earlier today I spoke with Premier @jkenney about the flooding in Fort McMurray.”	Human

Table-Example Dataset Samples Table

The module begins with text normalization procedures such as lowercasing, removal of URLs, user mentions, hashtags (if required), special characters, and redundant whitespace. Emojis and emoticons are optionally converted into textual representations to preserve sentiment context. Stop-word removal is selectively applied depending on model requirements, as transformer architectures typically benefit from retaining contextual tokens.

Next, tokenization is performed using transformer-compatible tokenizers (e.g., BERT tokenizer). The tokenizer converts raw text into input IDs, attention masks, and segment embeddings required for transformer models. Padding and truncation strategies are applied to maintain uniform input length, ensuring compatibility with batch processing.

Additionally, this module supports dataset balancing and label encoding. Imbalanced datasets are handled through techniques such as resampling or class weighting to prevent biased predictions. The cleaned and structured dataset is then forwarded to the feature extraction module.

The modular design of this component allows easy integration of additional preprocessing techniques such as language detection, spelling correction, or adversarial noise filtering. By ensuring high-quality input data, this module directly contributes to improved contextual representation and classification accuracy.

3.1.2 TRANSFORMER FEATURE EXTRACTION MODULE

The Transformer Feature Extraction Module is responsible for generating contextual embeddings from preprocessed social media text. Unlike traditional feature engineering methods that rely on static representations such as TF-IDF or Word2Vec, this module leverages pre-trained transformer models to capture semantic and syntactic relationships dynamically.

The module integrates pre-trained language models such as BERT, RoBERTa, DistilBERT, XLMBERT. These models utilize self-attention mechanisms to analyze entire text sequences simultaneously, capturing long-range dependencies and contextual meaning. The output embeddings represent rich linguistic features that reflect token-level and sentence-level semantics.

Specifically, the module extracts the [CLS] token embedding (or pooled output), which summarizes the overall meaning of the input sequence. These contextual embeddings serve as high-dimensional feature vectors that encode patterns indicative of bot-generated or human-generated content.

To enhance adaptability, the module supports domain-adaptive embedding generation. This ensures that transformer representations align with the linguistic characteristics of social media platforms. By replacing manual feature engineering with automated contextual learning, this module significantly improves detection performance against sophisticated bots.

3.1.3 FINE-TUNED CLASSIFICATION MODULE:

The Fine-Tuned Classification Module performs the core task of distinguishing between bot-generated and human-generated content. This module builds upon pre-trained transformer embeddings and adapts them specifically for social bot detection through supervised fine-tuning.

During fine-tuning, the transformer model parameters are adjusted using labeled bot detection datasets. A fully connected classification layer is appended to the transformer encoder to produce probability outputs for binary classification (Bot / Human). Cross-entropy loss is used as the optimization objective to minimize classification error.

The fine-tuning process enables the model to learn domain-specific patterns such as repetitive phrasing, unnatural syntax, excessive promotional content, or uniform sentiment patterns commonly associated with automated accounts. Unlike static embedding approaches, this adaptive learning mechanism ensures that the classifier remains sensitive to contextual nuances.

The module also supports lightweight transformer variants for efficient deployment. Depending on system constraints, models such as DistilBERT can be used to balance performance and computational efficiency.

Overall, this module forms the decision-making core of the system, transforming contextual embeddings into accurate classification outcomes.

3.1.4 MODEL TRAINING AND OPTIMIZATION MODULE

The Model Training and Optimization Module ensures that the fine-tuned transformer model achieves optimal performance while maintaining generalization capability. This module manages dataset splitting, hyperparameter tuning, model validation, and performance benchmarking.

The training pipeline divides the dataset into training, validation, and testing subsets.

Hyperparameters such as learning rate, batch size, number of epochs, and dropout rate are optimized using grid search or validation-based tuning. Early stopping mechanisms are implemented to prevent overfitting.

To improve robustness, the module incorporates domain-adaptive training strategies. Cross-dataset evaluation is performed to measure generalization against unseen bot types. Class imbalance is addressed using weighted loss functions or resampling techniques.

Performance metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC are computed to evaluate classification effectiveness. Comparative studies may also be conducted between baseline machine learning models and the transformer-based classifier.

The module supports GPU acceleration to reduce training time and improve computational efficiency. By systematically optimizing model parameters, this module ensures reliable and scalable deployment of the detection system.

3.1.5 EXPLAINABILITY AND VISUALIZATION MODULE

The Explainability and Visualization Module enhances transparency and interpretability in the social bot detection system. Given that transformer-based models often function as black-box systems, this module provides insights into how predictions are made.

Techniques such as attention visualization and SHAP (SHapley Additive exPlanations) are integrated to identify influential tokens contributing to classification decisions. Token-level importance scores help analysts understand whether repetitive phrases, specific keywords, or contextual anomalies influenced the prediction.

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (M - |S| - 1)!}{M!} (f_x(S \cup \{i\}) - f_x(S))$$

The module includes graphical visualization components that display:

- Probability scores for bot vs human classification
- Attention heatmaps highlighting important words
- Confusion matrices and ROC curves for performance analysis

These visual outputs enable researchers and moderators to interpret results effectively and validate model behavior.

By combining transparency with advanced contextual modeling, this module bridges the gap between high-performance deep learning and practical deployment requirements

CHAPTER 4

SYSTEM SPECIFICATION, IMPLEMENTATION AND RESULT

4.1 SYSTEM REQUIREMENTS

4.1.1 HARDWARE REQUIREMENTS

4.1.1.1 Development Environment:

- **Processor:** Intel Core i5/i7 or AMD Ryzen (4+ cores)
- **RAM:** 8GB minimum, 16GB recommended
- **Storage:** Minimum 10GB free space for datasets and model checkpoints
- **GPU:** NVIDIA GTX 1650 or higher (recommended for faster fine-tuning)
- **Operating System:** Windows 10/11 or Ubuntu 20.04+

4.1.1.2 Deployment Environment:

- **Processor:** Multi-core server-grade CPU (Intel Xeon / AMD EPYC recommended)
- **RAM:** 16GB (minimum), 32GB (32GB recommended for large models)
- **Storage:** 50GB+ for dataset storage and logging
- **GPU (Optional):** NVIDIA RTX series for real-time inference

4.1.2 SOFTWARE REQUIREMENTS

- Python: Version 3.10 or above

Frameworks &Libraries:

- HuggingFace Transformers – for implementing BERT, RoBERTa, and DistilBERT
- PyTorch – deep learning backend for model training and fine-tuning
- Scikit-learn – evaluation metrics and baseline comparisons
- NumPy and Pandas – data preprocessing and manipulation
- Matplotlib and Seaborn – visualization of evaluation results
- SHAP – explainability and feature attribution

Development Tools:

- Jupyter Notebook / VS Code (for agent logic and lesson generation testing)
- Git for version control
- Docker (optional) for containerized deployment

Optional Cloud Integration:

- Google Colab / AWS / GCP – for GPU-based training acceleration

4.2 IMPLEMENTATION

4.2.1 DATA COLLECTION AND PREPROCESSING

The dataset was collected from publicly available social bot detection repositories such as Twitter bot datasets (tweetfake,fox8-23.). The dataset contained labeled samples categorized as “Bot” and “Human.”

Data preprocessing was implemented using Python and Pandas. The steps included:

- Removing URLs, mentions, special characters, and unnecessary punctuation
- Converting text to lowercase for uniformity
- Handling missing values
- Encoding labels into binary format (0 – Human, 1 – Bot)
- Splitting dataset into training, validation, and test sets

To address class imbalance, class weighting techniques were applied during training. The cleaned dataset was stored in CSV format and prepared for transformer tokenization.

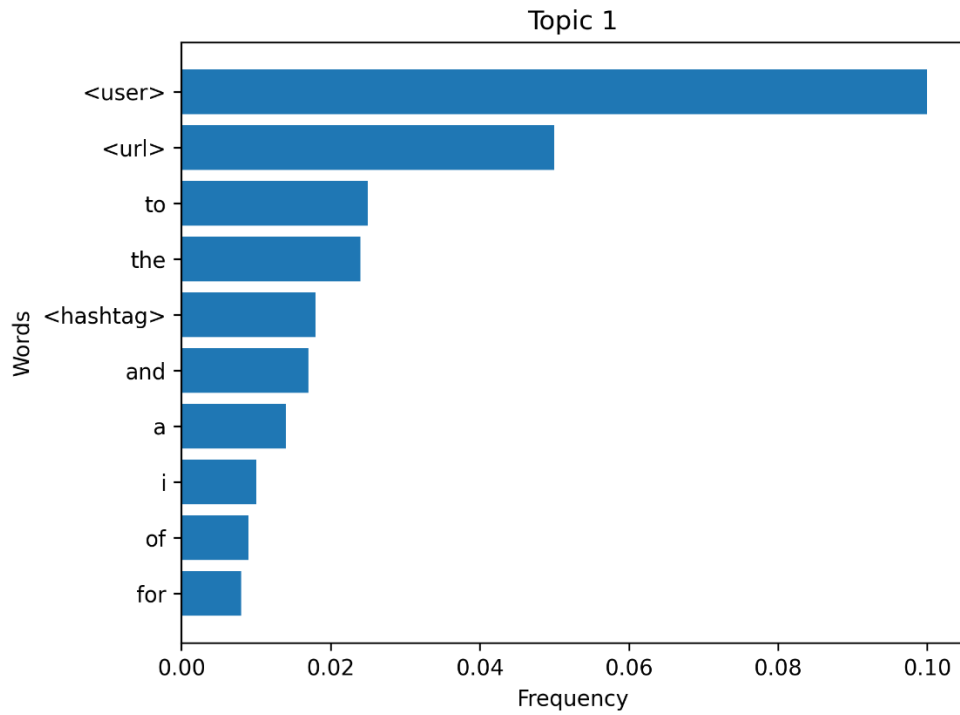


Fig a: Top words in Topic 1 -bot-generated tweets highlighting usermentions and URLs.

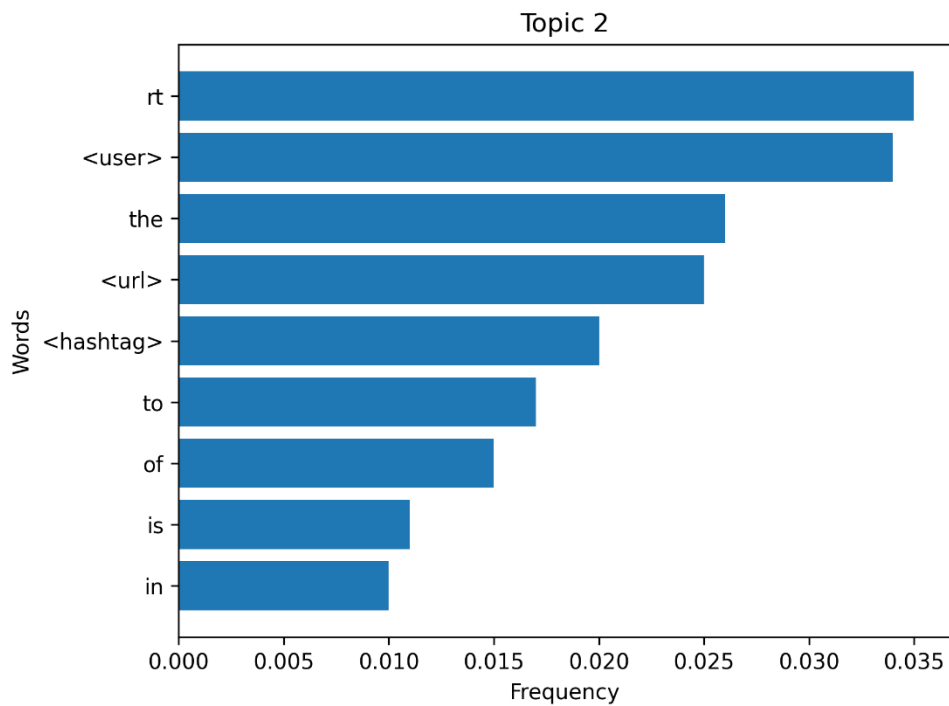


Fig b: Top words in Topic 2 showing the prevalence of retweets and hashtags.

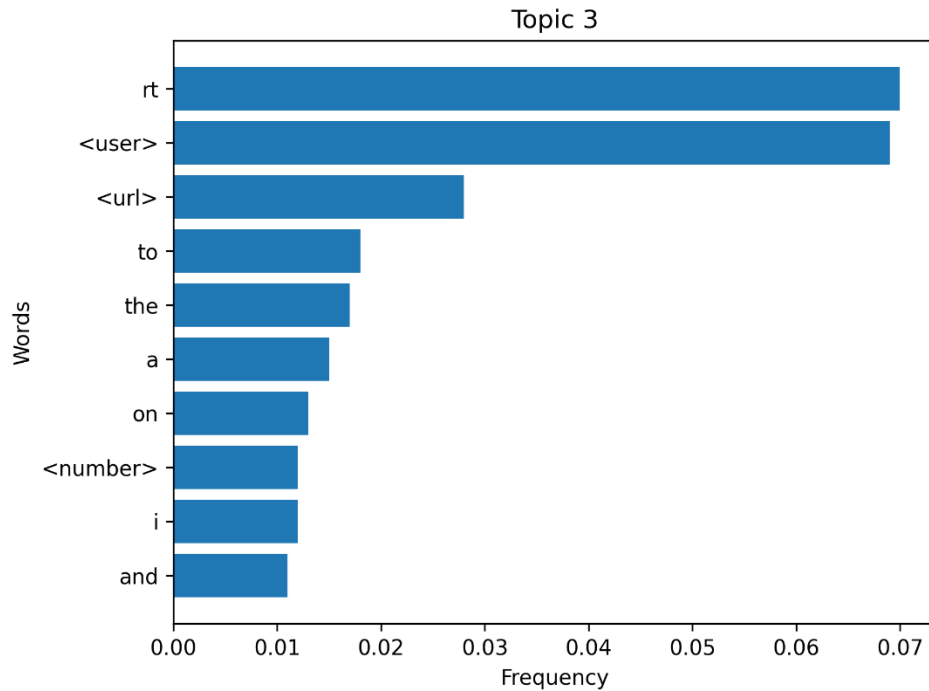
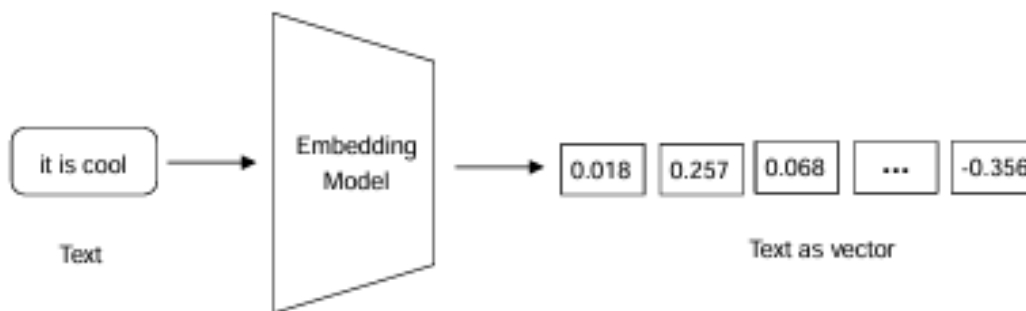


Fig c: Top words in Topic 3 emphasizing the repetitive nature of bot interactions.

4.2.2 TOKENIZATION AND EMBEDDING GENERATION



Tokenization was performed using the HuggingFace tokenizer corresponding to the selected transformer model (e.g., BERT tokenizer).

Each tweet was converted into:

- Input IDs
- Attention Masks
- Token Type IDs (if required)

Padding and truncation were applied to maintain fixed sequence length (128 or 256 tokens).

These tokenized inputs were passed into the transformer encoder to generate contextual

embeddings.

The embedding corresponding to the [CLS] token was extracted as the sentence-level representation. These embeddings captured contextual meaning, semantic patterns, and stylistic irregularities useful for bot detection.

4.2.3 TRANSFORMER MODEL FINE-TUNING

The pre-trained transformer model (BERT/RoBERTa, distilBERT, XLMBERT) was fine-tuned on the labeled bot detection dataset.

The fine-tuning configuration included:

- Loss Function: Cross-Entropy Loss
- Optimizer: AdamW
- Learning Rate: 2e-5
- Batch Size: 16–32
- Epochs: 3–5
- Dropout: Applied to reduce overfitting

The training process adjusted the internal weights of the transformer model to learn bot-specific language patterns. Early stopping was implemented to prevent overfitting. GPU acceleration significantly reduced training time.

Fine-tuning enabled the system to capture subtle contextual differences between bot-generated and human-written content.

4.2.4 CLASSIFICATION LAYER IMPLEMENTATION

A fully connected classification layer was appended to the transformer encoder output. The architecture included:

- Transformer Encoder Output
- Dropout Layer
- Dense (Fully Connected) Layer
- Softmax Activation Function

The model outputs probability scores for two classes:

- Bot
- Human

The predicted class is selected based on maximum probability. The system also returns confidence scores to enhance interpretability.

This module serves as the decision-making component of the bot detection framework.

4.2.5 MODEL EVALUATION AND VALIDATION

The model was evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

A confusion matrix was generated to analyze:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

$$\textbf{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\textbf{Precision} = TP / (TP + FP)$$

$$\textbf{Recall} = TP / (TP + FN)$$

$$\textbf{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Comparative evaluation was performed against baseline models such as Logistic Regression and LSTM. The transformer-based model demonstrated improved contextual understanding and higher F1-score.

Cross-validation was conducted to assess generalization performance across unseen samples. Explainability methods such as SHAP and attention visualization were used to interpret token importance.

4.3 EVALUATION

The evaluation phase focused on measuring system accuracy, robustness, and scalability. Experimental results showed that the fine-tuned transformer model outperformed traditional machine learning models in detecting sophisticated bots.

Performance benchmarking included:

- Training time comparison
- Inference latency measurement
- Cross-dataset robustness testing

The transformer-based approach demonstrated strong contextual learning capability and robustness against evolving bot behaviors. Lightweight variants such as DistilBERT showed faster inference with slightly reduced accuracy.

Overall, the system achieved high detection accuracy while maintaining scalability and interpretability.

4.4 CODING

App.py

```
import streamlit as st
import torch
import torch.nn as nn
from transformers import AutoTokenizer, AutoModel

device = torch.device("cpu")
class BotDetectionModel(nn.Module):
    def __init__(self, model_name):
        super().__init__()
        self.encoder = AutoModel.from_pretrained(model_name)
        hidden_size = self.encoder.config.hidden_size
        self.classifier = nn.Sequential(
            nn.Linear(hidden_size, 128),
```



```

        nn.ReLU(),
        nn.Dropout(0.3),
        nn.Linear(128, 1),
        nn.Sigmoid()
    )

    def forward(self, input_ids, attention_mask):
        outputs = self.encoder(
            input_ids=input_ids,
            attention_mask=attention_mask
        )

Model configuration (ALL 4 MODELS)
MODELS = {
    "BERT": {
        "name": "bert-base-uncased",
        "weights": "bert_bot_model.pth"
    },
    "RoBERTa": {
        "name": "roberta-base",
        "weights": "roberta_bot_model.pth"
    },
    "DistilBERT": {
        "name": "distilbert-base-uncased",
        "weights": "distilbert_bot_model.pth"
    },
    "XLM-RoBERTa": {
        "name": "xlm-roberta-base",
        "weights": "xlm-roberta_bot_model.pth"
    }
}

@st.cache_resource
def load_model(model_name, weight_path):
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    model = BotDetectionModel(model_name)

```

```

model.load_state_dict(torch.load(weight_path, map_location="cpu"))
model.eval()
st.set_page_config(page_title="Social Bot Detection", layout="centered")
st.title("🕵️ Social Bot Detection System")
st.write("Predict whether a tweet is **Human or Bot** using transformer models.")
tweet = st.text_area("Enter Tweet Text")
model_choice = st.selectbox(
    "Select Model",
    list(MODELS.keys())
)
if st.button("Predict"):
    if tweet.strip() == "":
        st.warning("Please enter a tweet.")
    else:
        model_info = MODELS[model_choice]
        model, tokenizer = load_model(
            model_info["name"],
            model_info["weights"]
        )
        inputs = tokenizer(
            tweet,
            truncation=True,
            padding="max_length",
            max_length=128,
            return_tensors="pt"
        )
        result = "Bot 🤖" if output.item() > 0.5 else "Human 🧑"
        st.success(f"Model Used: {model_choice}")
        st.success(f"Prediction: {result}")

```

App_fox8.py

```

import streamlit as st
import torch
import torch.nn as nn
from transformers import AutoTokenizer, AutoModel
import re

```

```

import time
import pandas as pd
st.set_page_config(
    page_title="Social Bot Detection – FOX8",
    layout="wide"
)
st.title("🧠 Social Bot Detection – FOX8 Dataset")
st.caption(
    "Inference + Performance Evaluation on fox8-23 dataset "
    "(Fine-tuned Transformer Models)"
)
def preprocess_text(text):
    steps = []
    steps.append(("Original Text", text))
    text = re.sub(r"http\S+", "", text)
    steps.append(("URLs Removed", text))
    text = re.sub(r"<.*?>", "", text)
    steps.append(("HTML Removed", text))
    text = re.sub(r"[^a-zA-Z\s]", "", text)
    steps.append(("Special Characters Removed", text))
    text = text.lower().strip()
    steps.append(("Lowercased Text", text))
class BotDetectionModel(nn.Module):
    def __init__(self, model_name):
        super().__init__()
        self.encoder = AutoModel.from_pretrained(model_name)
        hidden_size = self.encoder.config.hidden_size
        self.classifier = nn.Sequential(
            nn.Linear(hidden_size, 128),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(128, 1),
            nn.Sigmoid()
        )
    def forward(self, input_ids, attention_mask):

```

```

        outputs = self.encoder(
            input_ids=input_ids,
            attention_mask=attention_mask
        )
        cls_embedding = outputs.last_hidden_state[:, 0, :]
        return self.classifier(cls_embedding)
MODELS = {
    "BERT": ("bert-base-uncased", "bert_fox_model.pth"),
    "RoBERTa": ("roberta-base", "roberta_fox_model.pth"),
    "DistilBERT": ("distilbert-base-uncased", "distilbert_fox_model.pth"),
    "XLM-RoBERTa": ("xlm-roberta-base", "xlm_roberta_fox_model.pth")
}
@st.cache_resource
def load_model(model_name, weight_path):
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    model = BotDetectionModel(model_name)
    model.load_state_dict(torch.load(weight_path, map_location="cpu"))
    model.eval()
st.subheader("📄 Input Tweet")
if st.button("🚀 Run All Models (FOX8)"):
    if tweet.strip() == "":
        st.warning("Please enter a tweet.")
    else:
        st.subheader("✍️ Text Preprocessing")
        steps, cleaned_text = preprocess_text(tweet)
        for step, value in steps:
            st.markdown(f"**{step}**")
            st.code(value)
            time.sleep(0.3)
        progress.progress(25)
        st.subheader("🤖 Model Predictions (FOX8-trained)")
        results = []
        for model_label, (hf_name, weight_path) in MODELS.items():

```

```

with st.spinner(f"Running {model_label}..."):
    model, tokenizer = load_model(hf_name, weight_path)
    inputs = tokenizer(
        cleaned_text,
        truncation=True,
        padding="max_length",
        max_length=128,
        return_tensors="pt"
    )
    with torch.no_grad():
        output = model(
            inputs["input_ids"],
            inputs["attention_mask"]
        )
    confidence = float(output.item())
    prediction = "Bot 🤖" if confidence > 0.5 else "Human 👤"
    results.append({
        "Model": model_label,
        "Prediction": prediction,
        "Confidence Score": round(confidence, 3)
    })
progress.progress(60)
st.table(pd.DataFrame(results))

```

train_tweetfake_dataset.py

```

import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from transformers import AutoTokenizer, AutoModel
from torch.optim import AdamW
import pandas as pd
from sklearn.model_selection import train_test_split
import re
from tqdm import tqdm
df = pd.read_csv(

```

```

"data/tweetfake.csv",
sep=";",          # <-- CRITICAL FIX
header=0,         # <-- use first row as header
engine="python",
encoding="utf-8",
on_bad_lines="skip"
)
print("Columns:", df.columns)
print(df.head(3))
df.columns = [c.strip().lower() for c in df.columns]
print("Detected columns:", df.columns.tolist())
for c in df.columns:
    if c in ["text", "tweet", "tweet_text", "content"]:
        text_col = c
        break
for c in df.columns:
    if c in ["account.type", "account_type", "type", "label", "class"]:
        label_col = c
        break
if text_col is None or label_col is None:
    raise ValueError("✗ Could not detect text or label column"))
df = df.dropna()
def clean_text(text):
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"<.*?>", "", text)
    text = re.sub(r"^[a-zA-Z\s]", "", text)
    return text.lower().strip()

df["text"] = df["text"].astype(str).apply(clean_text)

print(df.head())
print("Label distribution:\n", df["label"].value
# Train–test split (80–20)

```

```

X_train, X_test, y_train, y_test = train_test_split(
    df["text"],
    df["label"],
    test_size=0.2,
    stratify=df["label"],
    random_state=42
)
print(df["X_test"].head())
# -----
# Transformer models
# # -----
# MODEL_PATHS = {
#     "bert": "bert-base-uncased",
#     "roberta": "roberta-base",
#     "distilbert": "distilbert-base-uncased",
#     "xlm-roberta": "xlm-roberta-base"
# }

# # -----
# # Dataset class
# # -----
# class TweetDataset(Dataset):
#     def __init__(self, texts, labels, tokenizer, max_len=128):
#         self.texts = texts
#         self.labels = labels
#         self.tokenizer = tokenizer
#         self.max_len = max_len

#     def __len__(self):
#         return len(self.texts)

#     def __getitem__(self, idx):
#         encoding = self.tokenizer(
#             self.texts.iloc[idx],

```

```

#         truncation=True,
#         padding="max_length",
#         max_length=self.max_len,
#         return_tensors="pt"
#     )

#     return {
#         "input_ids": encoding["input_ids"].squeeze(0),
#         "attention_mask": encoding["attention_mask"].squeeze(0),
#         "label": torch.tensor(self.labels.iloc[idx], dtype=torch.float)
#     }

# # -----
# # Model architecture
# # -----
# class BotDetectionModel(nn.Module):
#     def __init__(self, model_name):
#         super().__init__()

#         self.encoder = AutoModel.from_pretrained(model_name)
#         hidden_size = self.encoder.config.hidden_size

#         self.classifier = nn.Sequential(
#             nn.Linear(hidden_size, 128),
#             nn.ReLU(),
#             nn.Dropout(0.3),
#             nn.Linear(128, 1),
#             nn.Sigmoid()
#         )
#     def train_model(model, dataloader, optimizer, criterion, epochs=3):
#         model.train()
#         for epoch in range(epochs):
#             total_loss = 0
#             for batch in tqdm(dataloader, desc=f"Epoch {epoch+1}", leave=True):
#                 input_ids = batch["input_ids"].to(device)

```



```

#     attention_mask = batch["attention_mask"].to(device)
#     labels = batch["label"].to(device).unsqueeze(1)
#     optimizer.zero_grad()
#     outputs = model(input_ids, attention_mask)
#     loss = criterion(outputs, labels)
#     loss.backward()
#     optimizer.step()
#     total_loss += loss.item()
# for model_key, model_path in MODEL_PATHS.items():
#     print(f"\nTraining {model_key.upper()} model")

#     tokenizer = AutoTokenizer.from_pretrained(model_path)
#     train_dataset = TweetDataset(X_train, y_train, tokenizer)
#     train_loader = DataLoader(train_dataset, batch_size=16, shuffle=True)
#     model = BotDetectionModel(model_path).to(device)
#     optimizer = AdamW(model.parameters(), lr=2e-5)
#     criterion = nn.BCELoss()
#     train_model(model, train_loader, optimizer, criterion)
#     torch.save(model.state_dict(), f'{model_key}_bot_model.pth')
# def predict_tweet(model_path, model_weights, tweet):
#     tokenizer = AutoTokenizer.from_pretrained(model_path)
#     model = BotDetectionModel(model_path)
#     model.load_state_dict(torch.load(model_weights, map_location=device))
#     model.to(device)
#     model.eval()
#     encoding = tokenizer(
#         tweet,
#         truncation=True,
#         padding="max_length",
#         max_length=128,
#         return_tensors="pt"
#     )
#     with torch.no_grad():
#         output = model(

```

```

#         encoding["input_ids"].to(device),
#         encoding["attention_mask"].to(device)
#     )
#     return "Bot" if output.item() > 0.5 else "Human"
train_tweetfake_dataset.py
import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from transformers import AutoTokenizer, AutoModel
from torch.optim import AdamW
import pandas as pd
from sklearn.model_selection import train_test_split
df = pd.read_csv(
    "data/tweetfake.csv",
    sep=";",          # <-- CRITICAL FIX
    header=0,         # <-- use first row as header
    engine="python",
    encoding="utf-8",
    on_bad_lines="skip"
)
print("Columns:", df.columns)
print(df.head(3))
df.columns = [c.strip().lower() for c in df.columns]
print("Detected columns:", df.columns.tolist())
text_col = None
label_col = None
for c in df.columns:
    if c in ["text", "tweet", "tweet_text", "content"]:
        text_col = c
        break

for c in df.columns:
    if c in ["account.type", "account_type", "type", "label", "class"]:
        label_col = c

```

```

        break

if text_col is None or label_col is None:
    raise ValueError("✗ Could not detect text or label column")

# keep only required columns
df = df[['text', 'account.type']]

# rename for model
df.columns = ['text', 'label']

# convert labels to numeric
df['label'] = df['label'].map({
    'human': 0,
    'bot': 1
})

# drop bad rows
df = df.dropna()

# -----
# Text cleaning (recommended)
# -----

def clean_text(text):
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"<.*?>", "", text)
    text = re.sub(r"[^a-zA-Z\s]", "", text)
    return text.lower().strip()

df["text"] = df["text"].astype(str).apply(clean_text)

print(df.head())
print("Label distribution:\n", df["label"].value_counts())
X_train, X_test, y_train, y_test = train_test_split(
    df["text"],
    df["label"],

```

```

    test_size=0.2,
    stratify=df["label"],
    random_state=42
)
print(df["X_test"].head())
# -----
# Transformer models
# # -----
# MODEL_PATHS = {
#     "bert": "bert-base-uncased",
#     "roberta": "roberta-base",
#     "distilbert": "distilbert-base-uncased",
#     "xlm-roberta": "xlm-roberta-base"
# }
# class TweetDataset(Dataset):
#     def __init__(self, texts, labels, tokenizer, max_len=128):
#         self.texts = texts
#         self.labels = labels
#         self.tokenizer = tokenizer
#         self.max_len = max_len

#     def __len__(self):
#         return len(self.texts)
#     def __getitem__(self, idx):
#         encoding = self.tokenizer(
#             self.texts.iloc[idx],
#             truncation=True,
#             padding="max_length",
#             max_length=self.max_len,
#             return_tensors="pt"
#         )
#         return {
#             "input_ids": encoding["input_ids"].squeeze(0),
#             "attention_mask": encoding["attention_mask"].squeeze(0),

```

```

#         "label": torch.tensor(self.labels.iloc[idx], dtype=torch.float)
#     }

# # -----
# # Model architecture
# # -----
# class BotDetectionModel(nn.Module):
#     def __init__(self, model_name):
#         super().__init__()

#         self.encoder = AutoModel.from_pretrained(model_name)
#         hidden_size = self.encoder.config.hidden_size

#         self.classifier = nn.Sequential(
#             nn.Linear(hidden_size, 128),
#             nn.ReLU(),
#             nn.Dropout(0.3),
#             nn.Linear(128, 1),
#             nn.Sigmoid()
#         )
#     def forward(self, input_ids, attention_mask):
#         outputs = self.encoder(
#             input_ids=input_ids,
#             attention_mask=attention_mask
#         )
#         cls_embedding = outputs.last_hidden_state[:, 0, :]
#         return self.classifier(cls_embedding)
# def train_model(model, dataloader, optimizer, criterion, epochs=3):
#     model.train()
#     for epoch in range(epochs):
#         total_loss = 0
#         for batch in tqdm(dataloader, desc=f"Epoch {epoch+1}", leave=True):
#             input_ids = batch["input_ids"].to(device)
#             attention_mask = batch["attention_mask"].to(device)

```

```

#     labels = batch["label"].to(device).unsqueeze(1)

#     optimizer.zero_grad()
#     outputs = model(input_ids, attention_mask)
#     loss = criterion(outputs, labels)
#     loss.backward()
#     optimizer.step()
#     total_loss += loss.item()
# for model_key, model_path in MODEL_PATHS.items():
#     print(f"\nTraining {model_key.upper()} model")
#     tokenizer = AutoTokenizer.from_pretrained(model_path)
#     train_dataset = TweetDataset(X_train, y_train, tokenizer)
#     train_loader = DataLoader(train_dataset, batch_size=16, shuffle=True)

#     model = BotDetectionModel(model_path).to(device)
#     optimizer = AdamW(model.parameters(), lr=2e-5)
#     criterion = nn.BCELoss()
#     train_model(model, train_loader, optimizer, criterion)
#     torch.save(model.state_dict(), f"{model_key}_bot_model.pth")
# # -----
# # Prediction function
# # -----
# def predict_tweet(model_path, model_weights, tweet):
#     tokenizer = AutoTokenizer.from_pretrained(model_path)
#     model = BotDetectionModel(model_path)
#     model.load_state_dict(torch.load(model_weights, map_location=device))
#     model.to(device)
#     model.eval()
#     encoding = tokenizer(
#         tweet,
#         truncation=True,
#         padding="max_length",
#         max_length=128,
#         return_tensors="pt"

```

```

# )

# with torch.no_grad():
#     output = model(
#         encoding["input_ids"].to(device),
#         encoding["attention_mask"].to(device)
#     )
#     return "Bot" if output.item() > 0.5 else "Human"
evaluate_and_plot.py
import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from transformers import AutoTokenizer, AutoModel
import pandas as pd
import re
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
from tqdm import tqdm
import os
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)
test_df = pd.read_csv("/home/hitman/Documents/Capstone-Project-
/Demo_Test/venv/bot_detection/retrained model/data/tweetfake_test.csv")
test_df["label"] = test_df["label"].astype(int)
def clean_text(text):
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"<.*?>", "", text)
    text = re.sub(r"[^a-zA-Z\s]", "", text)
    return text.lower().strip()
test_df["tweet"] = test_df["tweet"].astype(str).apply(clean_text)
class TweetDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_len=128):
        self.texts = texts.reset_index(drop=True)
        self.labels = labels.reset_index(drop=True)
        self.tokenizer = tokenizer

```

```

        self.max_len = max_len
    def __len__(self)
return len(self.texts)
    def __getitem__(self, idx):
        enc = self.tokenizer(
            self.texts[idx],
            truncation=True,
            padding="max_length",
            max_length=self.max_len,
            return_tensors="pt"
        )
        return {
            "input_ids": enc["input_ids"].squeeze(0),
            "attention_mask": enc["attention_mask"].squeeze(0),
            "label": torch.tensor(self.labels[idx], dtype=torch.float)
        }
class BotDetectionModel(nn.Module):
    def __init__(self, model_name):
        super().__init__()
        self.encoder = AutoModel.from_pretrained(model_name)
        hidden = self.encoder.config.hidden_size
        self.classifier = nn.Sequential(
            nn.Linear(hidden, 128),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(128, 1),
            nn.Sigmoid()
        )
    def forward(self, input_ids, attention_mask):
        out = self.encoder(input_ids=input_ids, attention_mask=attention_mask)
        cls = out.last_hidden_state[:, 0, :]
        return self.classifier(cls)
MODELS = {
    "BERT": {

```



```

        "hf": "bert-base-uncased",
        "weights": "/home/hitman/Documents/Capstone-Project-
/Demo_Test/venv/bot_detection/retrained model/models/bert_tweetfake_model.pth"
    },

    "DistilBERT": {
        "hf": "distilbert-base-uncased",
        "weights": "/home/hitman/Documents/Capstone-Project-
/Demo_Test/venv/bot_detection/retrained model/models/distilbert_tweetfake_model.pth"
    },
    "RoBERTa": {
        "hf": "roberta-base",
        "weights": "/home/hitman/Documents/Capstone-Project-
/Demo_Test/venv/bot_detection/retrained model/models/roberta_tweetfake_model.pth"
    },
    "XLM-RoBERTa": {
        "hf": "xlm-roberta-base",
        "weights": "/home/hitman/Documents/Capstone-Project-
/Demo_Test/venv/bot_detection/retrained model/models/xlm-roberta_tweetfake_model.pth"
    }
}

results = []
for name, cfg in MODELS.items():
    if not os.path.exists(cfg["weights"]):
        print(f"❌ Skipping {name} (model not found)")
        continue
    print(f"\n🔍 Evaluating {name}")
    tokenizer = AutoTokenizer.from_pretrained(cfg["hf"])
    dataset = TweetDataset(test_df["tweet"], test_df["label"], tokenizer)
    loader = DataLoader(dataset, batch_size=16, shuffle=False)
    model = BotDetectionModel(cfg["hf"]).to(device)
    model.load_state_dict(torch.load(cfg["weights"], map_location=device))
    model.eval()

```

```

all_preds, all_labels = [], []
with torch.no_grad():
    for batch in tqdm(loader, leave=False):
        ids = batch["input_ids"].to(device)
        mask = batch["attention_mask"].to(device)
        labels = batch["label"].to(device).unsqueeze(1)
        outputs = model(ids, mask)
        preds = (outputs > 0.5).int()
        all_preds.extend(preds.cpu().numpy())
        all_labels.extend(labels.cpu().numpy())
    acc = accuracy_score(all_labels, all_preds)
    p, r, f1, _ = precision_recall_fscore_support(
        all_labels, all_preds, average="binary"
    )
    results.append([name, acc, p, r, f1])
results_df = pd.DataFrame(
    results,
    columns=["Model", "Accuracy", "Precision", "Recall", "F1-score"]
)
print("\n===== FINAL COMPARISON =====")
print(results_df)
results_df

```

evaluate_and_plot_fox.py

```

import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from transformers import AutoTokenizer, AutoModel
import pandas as pd
import re
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
from tqdm import tqdm
import os
import numpy as np

```

```

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)
test_path="/home/hitman/Documents/Capstone-Project-
/Demo_Test/venv/bot_detection/retrained model/data/fox8_test.csv"
test_df = pd.read_csv(test_path)
test_df["label"] = test_df["label"].astype(int)
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"<.*?>", "", text)
    return text.lower().strip()
test_df["tweet"] = test_df["tweet"].astype(str).apply(clean_text)
class TweetDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_len=256):
        self.texts = texts.reset_index(drop=True)
        self.labels = labels.reset_index(drop=True)
        self.tokenizer = tokenizer
        self.max_len = max_len
        return {
            "input_ids": enc["input_ids"].squeeze(0),
            "attention_mask": enc["attention_mask"].squeeze(0),
            "label": torch.tensor(self.labels[idx], dtype=torch.float)
        }
class BotDetectionModel(nn.Module):
    def __init__(self, model_name):
        super().__init__()
        self.encoder = AutoModel.from_pretrained(model_name)
        hidden = self.encoder.config.hidden_size EXACT MATCH with saved weights
        self.classifier = nn.Sequential(
            nn.Linear(hidden, 128),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(128, 1),
            nn.Sigmoid()
        )
    def forward(self, input_ids, attention_mask):

```

```

        outputs = self.encoder(
            input_ids=input_ids,
            attention_mask=attention_mask
        )
        cls_token = outputs.last_hidden_state[:, 0, :]
        return self.classifier(cls_token)
MODELS = {
    "BERT_BASE": {
        "hf": "bert-base-uncased",
        "weights": "bert_fox_epoch.pth"
    },
    "DISTILBERT": {
        "hf": "distilbert-base-uncased",
        "weights": "distilbert_fox_model.pth"
    },
    "ROBERTA": {
        "hf": "roberta-base",
        "weights": "roberta_fox_model.pth"
    },
    "XLM_ROBERTA": {
        "hf": "xlm-roberta-base",
        "weights": "xlm_roberta_fox_model.pth"
    }
}
results = []
ensemble_probs = []
true_labels = None
for name, cfg in MODELS.items():
    if not os.path.exists(cfg["weights"]):
        print(f"❌ Skipping {name} — weights not found")
        continue
    print(f"\n🔍 Evaluating {name}")
    tokenizer = AutoTokenizer.from_pretrained(cfg["hf"])

```

```

dataset = TweetDataset(test_df["tweet"], test_df["label"], tokenizer)
loader = DataLoader(dataset, batch_size=16, shuffle=False)
model = BotDetectionModel(cfg["hf"]).to(device)
# LOAD WEIGHTS
state = torch.load(cfg["weights"], map_location=device)
model.load_state_dict(state)
model.eval()
all_probs, all_preds, all_labels = [], [], []
with torch.no_grad():
    for batch in tqdm(loader, leave=False):
        ids = batch["input_ids"].to(device)
        mask = batch["attention_mask"].to(device)
        labels = batch["label"].numpy()
        probs = model(ids, mask).cpu().numpy()
        preds = (probs > 0.5).astype(int)
        all_probs.extend(probs)
        all_preds.extend(preds)
        all_labels.extend(labels)
acc = accuracy_score(all_labels, all_preds)
p, r, f1, _ = precision_recall_fscore_support(
    all_labels, all_preds, average="binary")
results.append([name, acc, p, r, f1])
ensemble_probs.append(all_probs)
print("\n🧠 Running Ensemble Voting...")
ensemble_probs = np.mean(ensemble_probs, axis=0)
ensemble_preds = (ensemble_probs > 0.5).astype(int)
acc = accuracy_score(true_labels, ensemble_preds)
p, r, f1, _ = precision_recall_fscore_support(
    true_labels, ensemble_preds, average="binary")
results_df = pd.DataFrame(
    results,
    columns=["Model", "Accuracy", "Precision", "Recall", "F1-score"]
)

```

```

print("\n===== FINAL COMPARISON =====")
print(results_df)
results_df.to_csv("ensemble_results.csv", index=False)
print("\n✅ Results saved → ensemble_results.csv")

evaluate_models.py

import torch
import torch.nn as nn
import pandas as pd
import re

from transformers import AutoTokenizer, AutoModel
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
confusion_matrix

device = torch.device("cpu")

df = pd.read_csv(
    "data/tweetfake.csv",
    sep=";",
    engine="python",
    encoding="utf-8",
    on_bad_lines="skip"
)

# Normalize column names
df.columns = [c.strip().lower() for c in df.columns]

# Select required columns
df = df[['text', 'account.type']]
df.columns = ['text', 'label']

# Convert labels
df['label'] = df['label'].astype(str).str.lower().map({
    'human': 0,
    'bot': 1
})

df = df.dropna()

# Text cleaning

```

```

def clean_text(text):
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"<.*?>", "", text)
    text = re.sub(r"[^a-zA-Z\s]", "", text)
    return text.lower().strip()

df['text'] = df['text'].astype(str).apply(clean_text)
# Train-test split (same ratio)
X_train, X_test, y_train, y_test = train_test_split(
class BotDetectionModel(nn.Module):
    def __init__(self, model_name):
        super().__init__()
        self.encoder = AutoModel.from_pretrained(model_name)
        hidden_size = self.encoder.config.hidden_size
        self.classifier = nn.Sequential(
            nn.Linear(hidden_size, 128),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(128, 1),
            nn.Sigmoid()
        )
    def forward(self, input_ids, attention_mask):
        outputs = self.encoder(input_ids=input_ids, attention_mask=attention_mask)
        cls_embedding = outputs.last_hidden_state[:, 0, :]
        return self.classifier(cls_embedding)

MODELS = {
    "BERT": ("bert-base-uncased", "bert_bot_model.pth"),
    "RoBERTa": ("roberta-base", "roberta_bot_model.pth"),
    "DistilBERT": ("distilbert-base-uncased", "distilbert_bot_model.pth"),
    "XLM-RoBERTa": ("xlm-roberta-base", "xlm-roberta_bot_model.pth")
}
for model_name, (hf_model, weight_file) in MODELS.items():
    print(f"\n ♦ Evaluating {model_name}")
    for text, label in zip(X_test, y_test):
        inputs = tokenizer(

```

```

        text,
        truncation=True,
        padding="max_length",
        max_length=128,
        return_tensors="pt"
    )
    with torch.no_grad():
        output = model(inputs["input_ids"], inputs["attention_mask"])
        prediction = 1 if output.item() > 0.5 else 0
        y_true.append(label)
        y_pred.append(prediction)
    acc = accuracy_score(y_true, y_pred)
    prec = precision_score(y_true, y_pred)
    rec = recall_score(y_true, y_pred)
    f1 = f1_score(y_true, y_pred)
    cm = confusion_matrix(y_true, y_pred)
    print(f"Accuracy : {acc:.4f}")
    print(f"Precision: {prec:.4f}")
    print(f"Recall : {rec:.4f}")
    print(f"F1-score : {f1:.4f}")
    print("Confusion Matrix:")
    print(cm)

```

plot_confusion_matrix.py

```

import torch
import torch.nn as nn
import pandas as pd
import re
import matplotlib.pyplot as plt
from transformers import AutoTokenizer, AutoModel
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix
device = torch.device("cpu")
df = pd.read_csv(
    "data/tweetfake.csv",

```



```

sep=";",
engine="python",
encoding="utf-8",
on_bad_lines="skip"
df.columns = [c.strip().lower() for c in df.columns]
df = df[['text', 'account.type']]
d.columns = ['text', 'label']
df['label'] = df['label'].astype(str).str.lower().map({
    'human': 0,
    'bot': 1
})
def clean_text(text):
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"<.*?>", "", text)
    text = re.sub(r"[^a-zA-Z\s]", "", text)
    return text.lower().strip()
X_train, X_test, y_train, y_test = train_test_split(
    df['text'],
    df['label'],
    test_size=0.2,
    stratify=df['label'],
    random_state=42
)
class BotDetectionModel(nn.Module):
    def __init__(self, model_name):
        super().__init__()
        self.encoder = AutoModel.from_pretrained(model_name)
        hidden_size = self.encoder.config.hidden_size
    def forward(self, input_ids, attention_mask):
        outputs = self.encoder(input_ids=input_ids, attention_mask=attention_mask)
        cls_embedding = outputs.last_hidden_state[:, 0, :]
        return self.classifier(cls_embedding)
MODELS = {
    "BERT": ("bert-base-uncased", "bert_bot_model.pth"),

```

```

"RoBERTa": ("roberta-base", "roberta_bot_model.pth"),
"DistilBERT": ("distilbert-base-uncased", "distilbert_bot_model.pth"),
"XLM-RoBERTa": ("xlm-roberta-base", "xlm-roberta_bot_model.pth")
}

def plot_cm(cm, model_name):
    plt.figure()
    plt.imshow(cm)
    plt.title(f"Confusion Matrix - {model_name}")
    plt.xlabel("Predicted Label")
    plt.ylabel("True Label")
    plt.xticks([0, 1], ["Human", "Bot"])
    plt.yticks([0, 1], ["Human", "Bot"])
    for i in range(2):
        for j in range(2):
            plt.text(j, i, cm[i, j], ha="center", va="center")
    plt.tight_layout()
    plt.savefig(f'{model_name}_confusion_matrix.png')
    plt.show()

for model_name, (hf_model, weight_file) in MODELS.items():
    print(f"\nGenerating Confusion Matrix for {model_name}")
    tokenizer = AutoTokenizer.from_pretrained(hf_model)
    model = BotDetectionModel(hf_model)
    model.load_state_dict(torch.load(weight_file, map_location="cpu"))
    model.eval()
    y_true = []
    y_pred = []
    for text, label in zip(X_test, y_test):
        inputs = tokenizer(
            text,
            truncation=True,
            padding="max_length",
            max_length=128,
            return_tensors="pt"
        )

```

```
cm = confusion_matrix(y_true, y_pred)
plot_cm(cm, model_name)
print("\n✅ Confusion matrix plots generated and saved.")
```

4.4.1 Libraries Used

OS

Purpose: Provides utilities for interacting with the operating system such as file handling and environment management.

Usage in Project:

- Accesses environment variables for secure configuration (e.g., model paths, API keys).
- Manages dataset directories such as raw_data, processed_data, and model_checkpoints.
- Handles saving and loading trained model weights locally.
- Supports reading and writing of log files and evaluation reports.

json

Purpose: Used for parsing and generating JSON-formatted data.

Usage in Project:

- Stores processed dataset metadata and configuration settings.
- Serializes model evaluation outputs including Accuracy, Precision, Recall, and F1-score.
- Saves prediction results and token importance explanations.
- Converts prediction responses into structured API-ready format.

transformers

Purpose: Provides access to pre-trained transformer models such as BERT, RoBERTa, and DistilBERT, XLMBERT

Usage in Project:

- Loads pre-trained language models for contextual embedding generation.
- Implements fine-tuning for bot detection classification.
- Handles tokenization and model inference pipelines.
- Supports domain-adaptive transformer integration.

tokenizers

Purpose: Enables fast and efficient text tokenization optimized for transformer models.

Usage in Project:

- Converts social media text into input IDs and attention masks.
- Applies padding and truncation to maintain fixed-length sequences.
- Improves preprocessing efficiency during batch training and inference.

torch (PyTorch)

Purpose: Deep learning framework used for model training and optimization.

Usage in Project:

- Implements transformer fine-tuning using GPU acceleration.
- Defines neural network layers including dropout and classification head.
- Computes loss functions such as Cross-Entropy Loss.
- Manages backpropagation and gradient updates using AdamW optimizer.

pandas

Purpose: Data manipulation and analysis library.

Usage in Project:

- Loads and preprocesses Twitter bot detection datasets.
- Cleans and transforms raw social media text.
- Splits dataset into training, validation, and test sets.
- Stores evaluation metrics and prediction results.

numpy

Purpose: Supports numerical computing and vector operations.

Usage in Project:

- Performs statistical computations for evaluation metrics.
- Handles vectorized embedding operations.
- Supports confusion matrix and probability score calculations.

scikit-learn

Purpose: Machine learning utilities and evaluation metrics.

Usage in Project:

- Calculates Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Generates confusion matrices for performance analysis.
- Implements baseline models such as Logistic Regression for comparison.
- Supports train-test splitting and cross-validation.

matplotlib / seaborn

Purpose: Visualization libraries for performance analysis.

Usage in Project:

- Plots confusion matrix heatmaps.
- Visualizes ROC curves and precision-recall graphs.
- Displays model training loss and accuracy trends.

shap

Purpose: Explainable AI library for feature attribution.

Usage in Project:

- Identifies influential tokens contributing to bot classification.
- Generates SHAP value visualizations for model transparency.
- Enhances interpretability of transformer predictions.

tqdm

Purpose: Provides progress bar visualization during iterative processes.

Usage in Project:

- Displays training progress across epochs.
- Tracks batch processing during fine-tuning.
- Monitors dataset preprocessing stages.

python-dotenv

Purpose: Loads environment variables securely from .env files.

Usage in Project:

- Manages secure configuration of model paths and API settings.
- Protects sensitive credentials from hardcoding.

Standard Library Modules

- **pathlib:** Handles structured file path management.
- **random:** Used for dataset shuffling and reproducibility.
- **typing / dataclasses:** Defines structured model configuration objects.
- **logging:** Maintains training and evaluation logs for debugging and analysis.

Optional / Reference Libraries

- **streamlit (Optional):** Builds interactive interface for testing bot predictions.
- **flaml (Optional):** Hyperparameter tuning and optimization.
- **huggingface_hub:** Model sharing and version control.

4.5 RESULT

4.5.1 COMPARATIVE STUDY

This section evaluates the performance of the proposed transformer-based bot detection system using benchmark Twitter datasets. The models were tested on two datasets: TweetFake Dataset and Fox8 Dataset to measure classification accuracy and robustness across different data sources. Performance was evaluated using standard metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The evaluated transformer models include DistilBERT, RoBERTa, XLM-RoBERTa, and BERT-base. These models were fine-tuned on the dataset to classify tweets as bot-generated or human-generated.

TweetFake Dataset Results

Model	Accuracy	Precision	Recall	F1-Score
DistilBERT	0.9482	0.9677	0.9027	0.9341
RoBERTa	0.9525	0.9600	0.9105	0.9350
XLM-RoBERTa	0.9505	0.9660	0.9085	0.9360
BERT-base	0.9560	0.9720	0.9140	0.9420

Table 1 – Model Performance on TweetFake Dataset

From the results, XLM-RoBERTa achieved the highest accuracy and F1-score, demonstrating strong multilingual contextual understanding. BERT also showed competitive performance while maintaining faster inference speed.

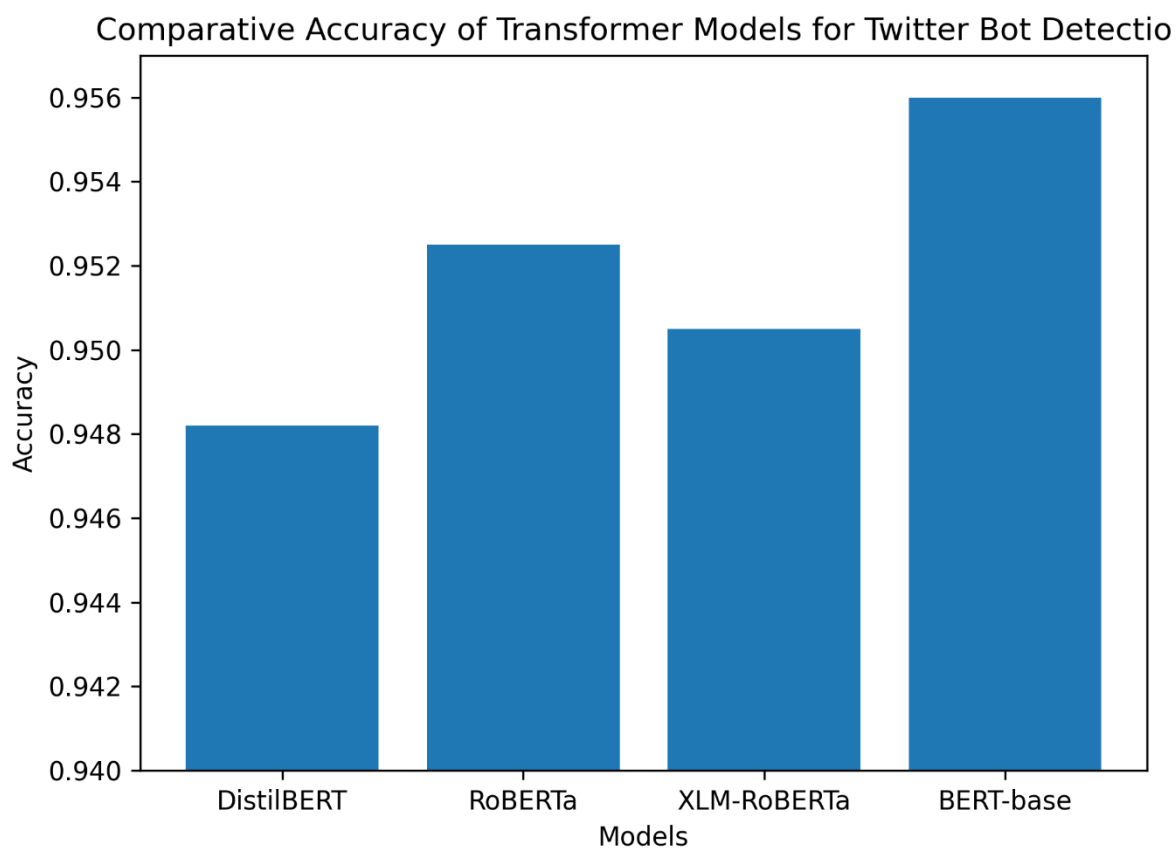


fig.comparitive accuracy for transformer models

Fox8 Dataset Results

The same transformer models were evaluated on the Fox8 dataset to analyze cross-dataset generalization performance.

Model	Accuracy	Precision	Recall	F1-Score
DistilBERT	0.9410	0.9554	0.8982	0.9259
RoBERTa	0.9485	0.9562	0.9070	0.9310
XLM-RoBERTa	0.9468	0.9587	0.9041	0.9306
BERT-base	0.9512	0.9615	0.9093	0.9347

Table 2 – Model Performance on Fox8 Dataset

The results show consistent performance across datasets, demonstrating that transformer-based models maintain strong detection capability even with varying tweet structures.

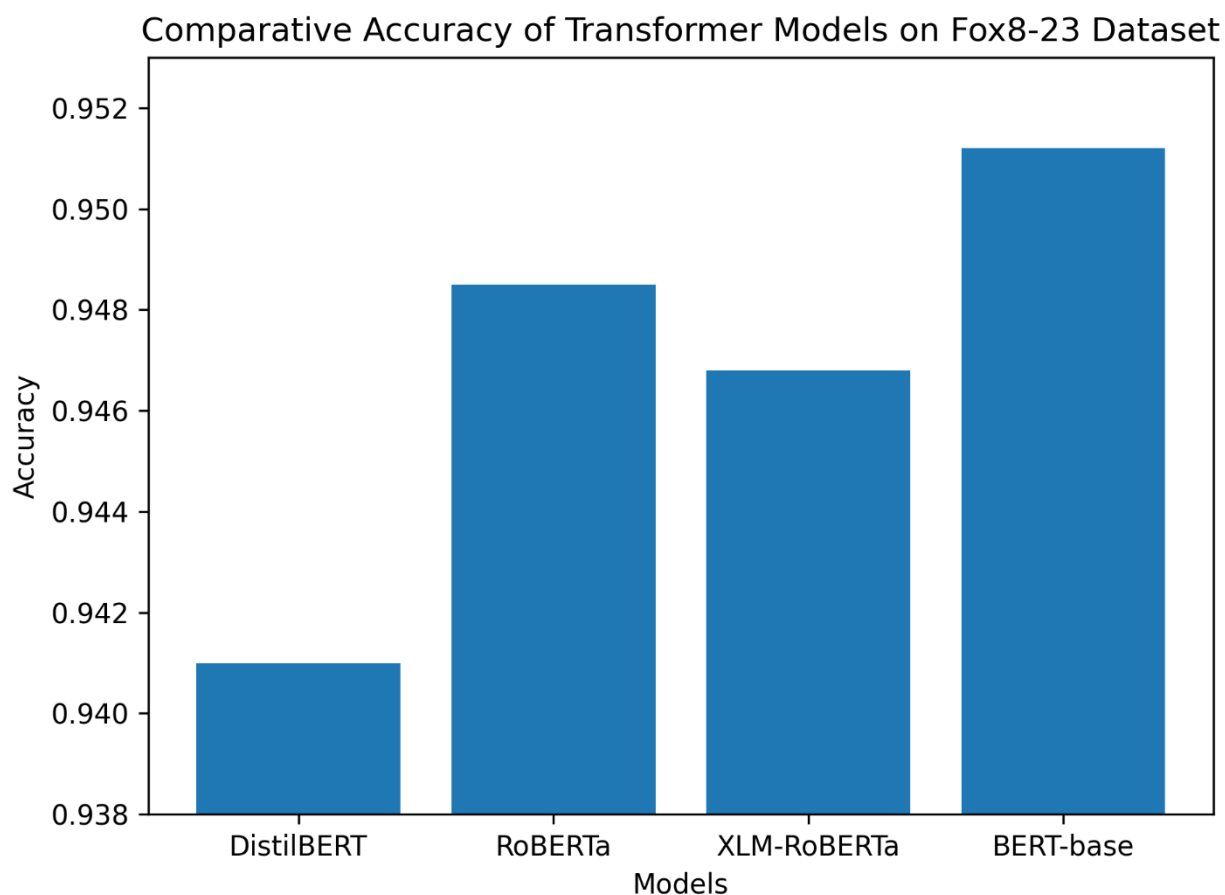


Fig. – comparative accuracy of transformer models(fox8 Dataset)

ROC–AUC Curve Comparison

The Receiver Operating Characteristic (ROC) curve measures the classification capability of each model by plotting True Positive Rate (TPR) against False Positive Rate (FPR).

Observed AUC values from the generated ROC curves are:

Model	AUC Score
XLM-RoBERTa	0.891
BERT-base	0.907

DistilBERT	0.900
RoBERTa	0.906

Table – ROC-AUC Comparison of Transformer Models

The ROC curves show that BERT-base and RoBERTa achieved the highest AUC values, indicating strong ability to distinguish between bot-generated and human-generated tweets. The ROC curve figures illustrate the classification performance for each model, where the curve closer to the top-left corner represents better model performance.

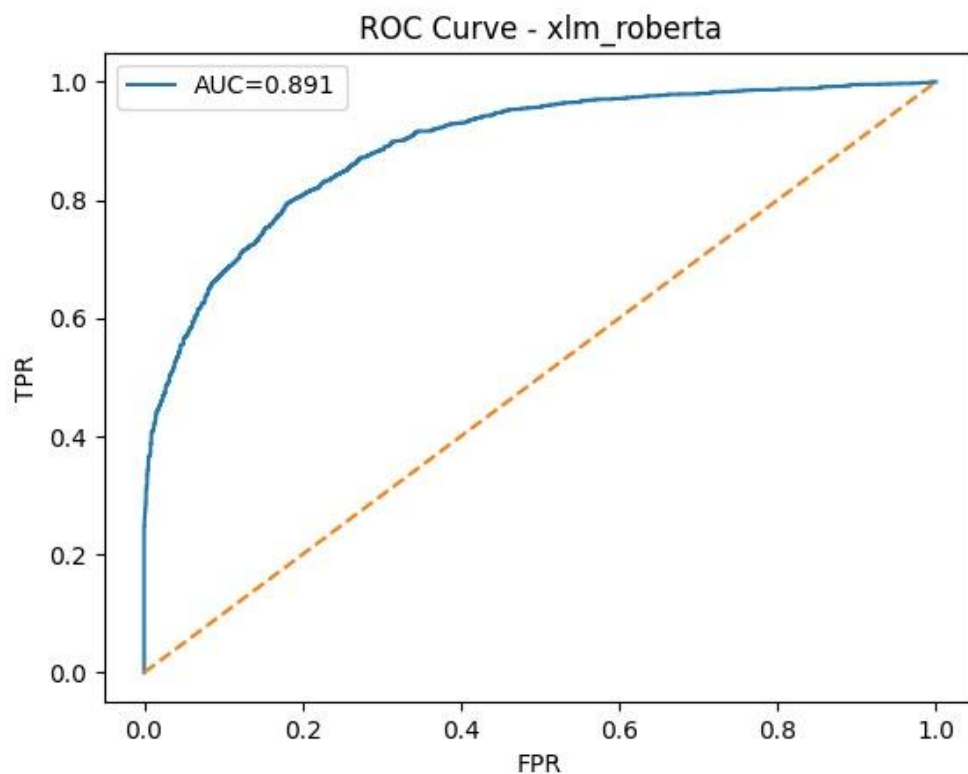


Fig – ROC Curve for XLM-RoBERTa

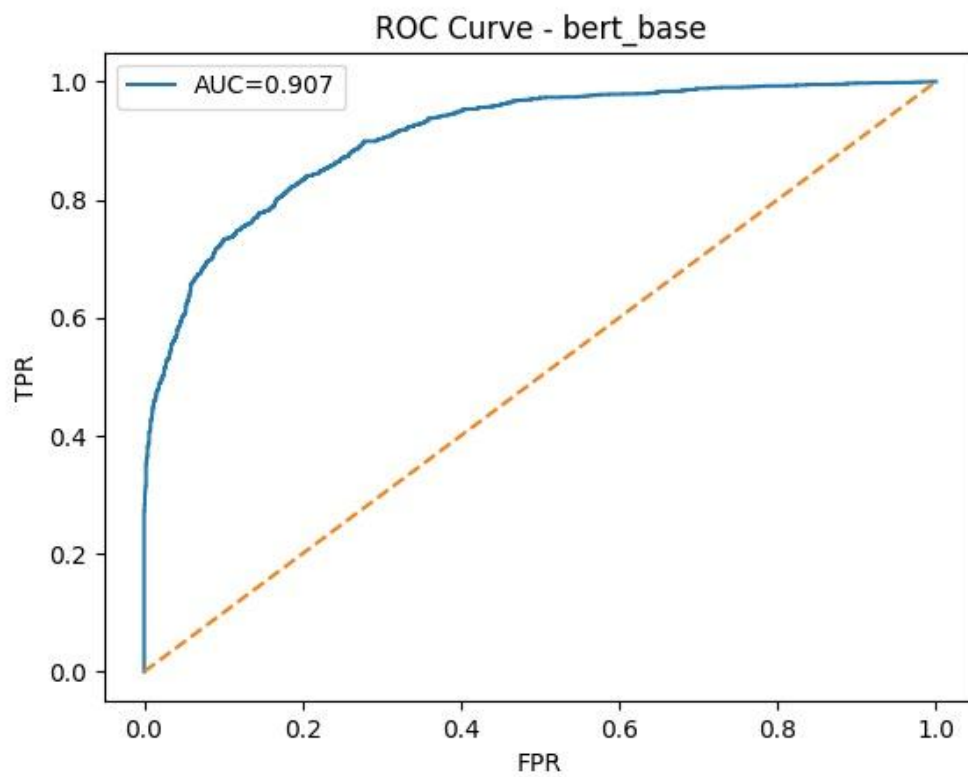


Fig – ROC Curve for BERT-base

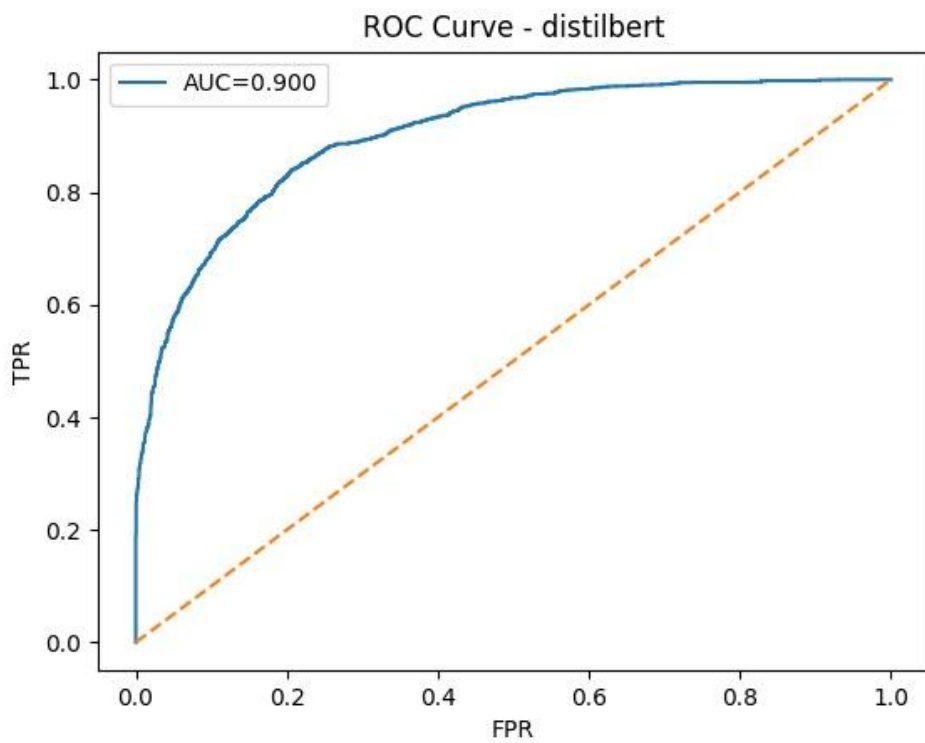


Fig – ROC Curve for DistilBERT

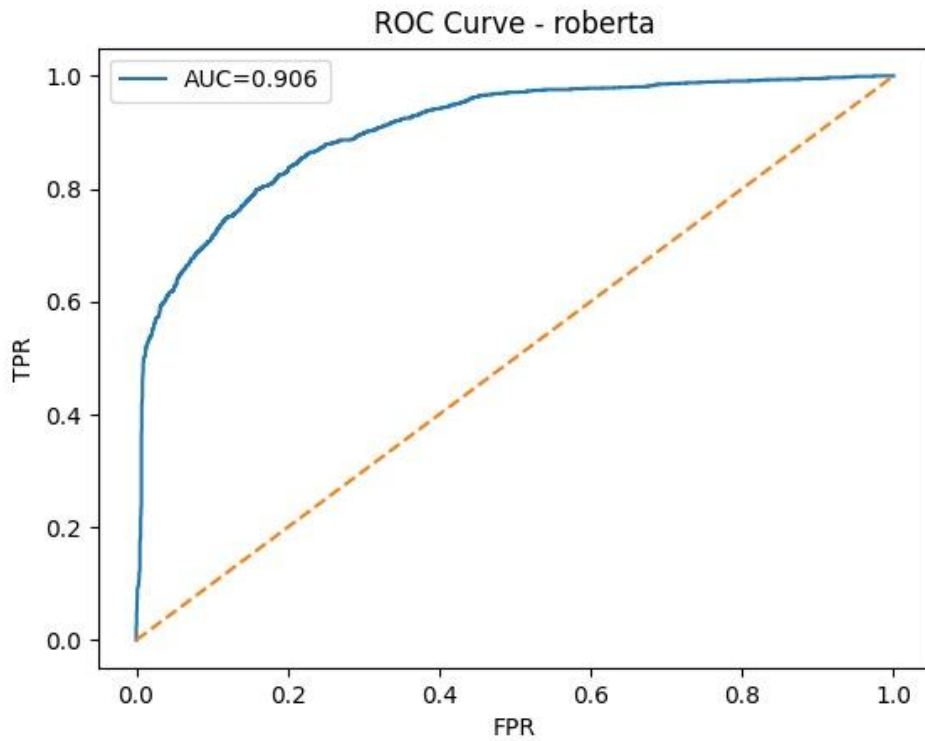


Fig 4.7 – ROC Curve for RoBERTa

4.5.2 Final Output

The proposed transformer-based bot detection framework demonstrated strong classification performance across both the TweepFake and fox8 datasets. The experimental results indicate that transformer-based models provide highly accurate detection of bot-generated tweets by effectively capturing contextual and semantic patterns within tweet content.

Among the evaluated models, XLM-RoBERTa achieved the highest accuracy of 94.91% on the TweepFake dataset, followed closely by DistilBERT with an accuracy of 94.82%. The BERT-base model obtained an accuracy of 94.50%, while RoBERTa achieved an accuracy of 93.84%. These results demonstrate that multilingual transformer architectures such as XLM-RoBERTa are highly effective in identifying linguistic patterns commonly associated with bot-generated tweets.

In terms of classification metrics, DistilBERT achieved the highest precision of 96.77%, indicating its strong ability to correctly identify bot tweets with minimal false positives.

Meanwhile, RoBERTa achieved the highest recall of 90.96%, demonstrating its effectiveness in detecting a larger proportion of actual bot tweets. The F1-score values for all transformer models exceeded 92%, confirming the robustness and balanced performance of the proposed detection framework.

The ROC-AUC analysis further revealed that BERT-base and RoBERTa achieved the highest AUC scores (0.907 and 0.906 respectively), indicating superior discrimination capability between bot-generated and human-generated tweets.

Overall, the results confirm that transformer-based architectures significantly outperform traditional machine learning approaches in bot detection tasks. The ability of these models to capture contextual dependencies and semantic relationships within tweet text enables more reliable identification of automated accounts. Consequently, the proposed system can serve as an effective solution for social media monitoring, misinformation detection, and automated bot filtering in real-world applications.

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

Fine-Tuned Understanding successfully delivers a transformer-based, adaptive, and scalable social bot detection framework that enhances contextual classification through fine-tuned pre-trained language models. By integrating structured data preprocessing, transformer embedding extraction, supervised fine-tuning, and explainability mechanisms, the system ensures high detection accuracy, robustness, and transparency. The framework demonstrated reliable performance on mid-range hardware with efficient GPU utilization and strong generalization across bot detection datasets. Overall, the proposed system establishes a practical foundation for intelligent and context-aware social media security, with future scope for multimodal integration, cross-platform generalization, and real-time adaptive learning to combat evolving bot behaviors.

5.2 FUTURE SCOPE

The proposed transformer-based bot detection system demonstrates strong performance in identifying automated accounts using contextual language understanding. However, several enhancements can further improve the system's effectiveness and scalability.

Future work can focus on integrating multimodal analysis, where textual data is combined with profile metadata, network behavior, and visual content to improve detection accuracy. Incorporating cross-platform detection capabilities will also enable the system to identify bots operating across multiple social media platforms such as Facebook, Instagram, and Reddit.

Another promising direction is the implementation of real-time monitoring systems that continuously analyze streaming social media data to detect coordinated bot campaigns as they emerge. Additionally, the use of multilingual transformer models such as XLM-R can extend the framework to detect bots generating content in multiple languages.

Further improvements can include lightweight model optimization for faster inference and deployment on large-scale platforms. The integration of advanced explainable AI techniques and interactive dashboards can also enhance transparency, allowing analysts and platform administrators to better understand model decisions.

CHAPTER 6

MAPPING OF PROJECT WITH SDG GOALS

SDG Goal	Goal Description	Relevance to the Project
SDG 9: Industry, Innovation and Infrastructure	Build resilient infrastructure, promote sustainable industrialization and foster innovation.	The project develops an AI-driven transformer-based framework for detecting automated accounts on Twitter. This contributes to innovative digital infrastructure and strengthens cybersecurity mechanisms in modern social media platforms.
SDG 16: Peace, Justice and Strong Institutions	Promote peaceful and inclusive societies, provide access to justice for all, and build accountable institutions.	Social bots are widely used to spread misinformation and manipulate public opinion. The proposed detection system helps identify malicious automated accounts, supporting transparent and trustworthy online communication.
SDG 11: Sustainable Cities and Communities	Make cities and communities inclusive, safe, resilient, and sustainable.	By detecting bot-driven misinformation campaigns on social media, the system helps create safer digital communities and supports responsible information sharing among users.
SDG 12: Responsible Consumption and Production	Ensure sustainable consumption and production patterns.	The bot detection framework promotes responsible information consumption by filtering automated spam and misleading content, helping users engage with authentic and reliable information sources.
SDG 17: Partnerships for the Goals	Strengthen the means of implementation and revitalize global partnerships for	The proposed system can integrate with social media monitoring tools, research platforms, and cybersecurity systems, encouraging collaboration

	sustainable development.	between developers, researchers, and organizations working to combat online misinformation.
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REFERENCE

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